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Efficient Evaluation of LLM Performance with Statistical Guarantees

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Before we begin, a huge thank you to the team



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Problem: LLM evaluation very important, but expensive.

- **Which model variant (of potentially thousands) should we deploy?**
  - **HuggingFace:** 1.86M models, 300K for text-generation
  - **Countless variants:** pre-training × post-training × distillation/quantization
- **Standard approach:** evaluate on **finite, curated benchmark suites**
  - **Expensive:** expert grading, long-form reasoning, multi-turn interactions
  - **Statistical challenge:** not representative of an i.i.d. super population!

1,000 Models × 10,000 Questions = A Lot of Experts' Time + $$$

Goal: Budget efficiency + statistical guarantees

- **Benchmark:** fixed, finite bank of N_q questions
- **New model:** correctness $z_j \in \{0,1\}$ on question j
- **History:** partially observed observations from N_{old} previous models

Target: finite-bank accuracy

$$\theta := \frac{1}{N_q} \sum_{j=1}^{N_q} z_j \in [0,1]$$

- **Statistical Task:** with only $n_b \ll N_q$ queries, construct the narrowest possible model-free frequentist-valid confidence interval (CI) for θ

Factorized Active Querying (FAQ) At-A-Glance

1. Learn Structure from History

Latent structure reveals which questions are informative

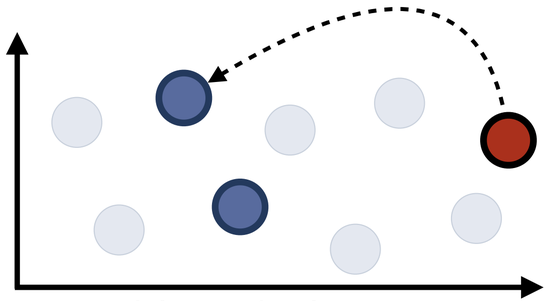
questions

models

1	0	?	1	?	0	1	1
?	1	0	1	0	?	1	0
0	?	1	?	1	0	0	1
1	1	?	0	1	1	?	0
?	0	1	1	0	?	1	?

2. Adaptive Querying

Spend budget on "most informative" questions



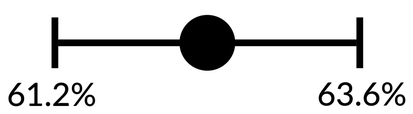
model-question latent space

Explore: learn new model factor
Exploit: shrink the CI

3. Report Accuracy + CI

Statistical validity despite non-traditional adaptive querying + strong dependencies

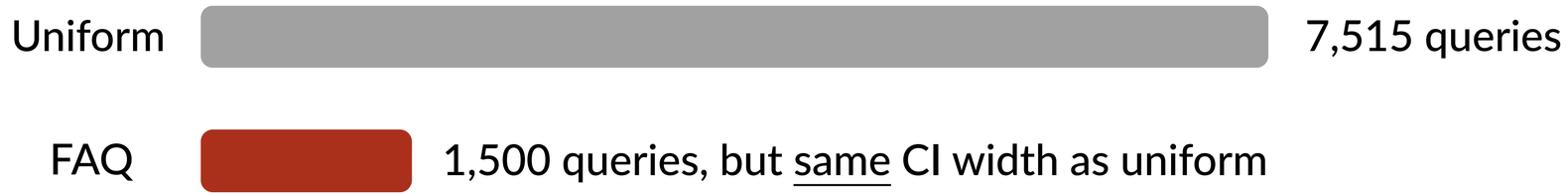
62.4%
Estimated benchmark accuracy



model-free
95% coverage

Same precision with far fewer model queries

MMLU-Pro across 2.2K models on HuggingFace's Open LLM Leaderboard



Up to 5x+

Effective Sample Size Gains,
≈95% empirical coverage

Factor models increases efficiency, while PAI preserves statistical validity

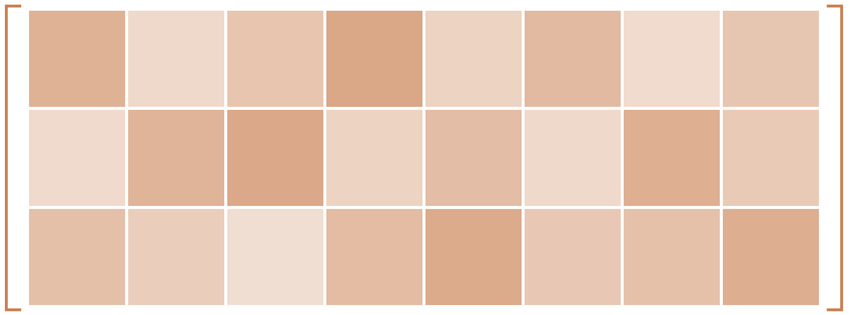
1. Learn structure from history: probabilistic factor models

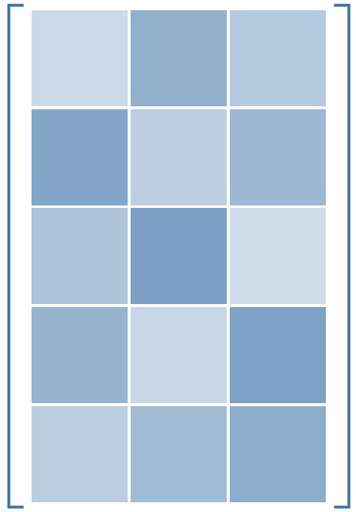
$$\begin{bmatrix} 1 & 0 & ? & 1 & ? & 0 & 1 & 1 \\ ? & 1 & 0 & 1 & 0 & ? & 1 & 0 \\ 0 & ? & 1 & ? & 1 & 0 & 0 & 1 \\ 1 & 1 & ? & 0 & 1 & 1 & ? & 0 \\ ? & 0 & 1 & 1 & 0 & ? & 1 & ? \end{bmatrix}$$

Historical Data H

1. Learn structure from history: probabilistic factor models

Question Factors $V^\top$





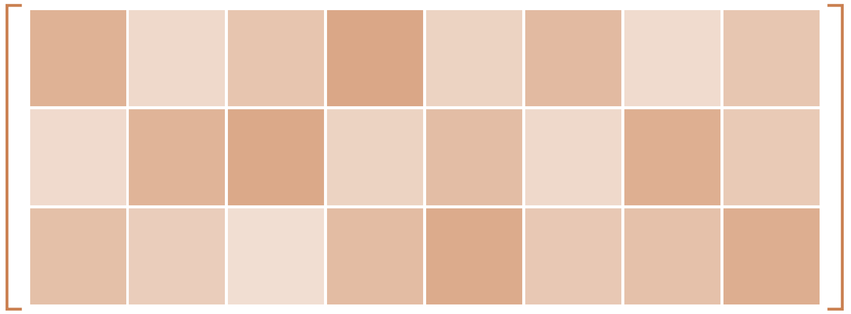
Model Factors U

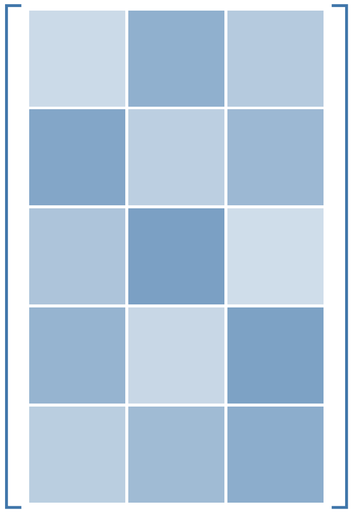
1	0	?	1	?	0	1	1
?	1	0	1	0	?	1	0
0	?	1	?	1	0	0	1
1	1	?	0	1	1	?	0
?	0	1	1	0	?	1	?

Historical Data H

1. Learn structure from history: probabilistic factor models

Question Factors $V^\top$





Model Factors U

1	0	?	1	?	0	1	1
?	1	0	1	0	?	1	0
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1	1	?	0	1	1	?	0
?	0	1	1	0	?	1	?

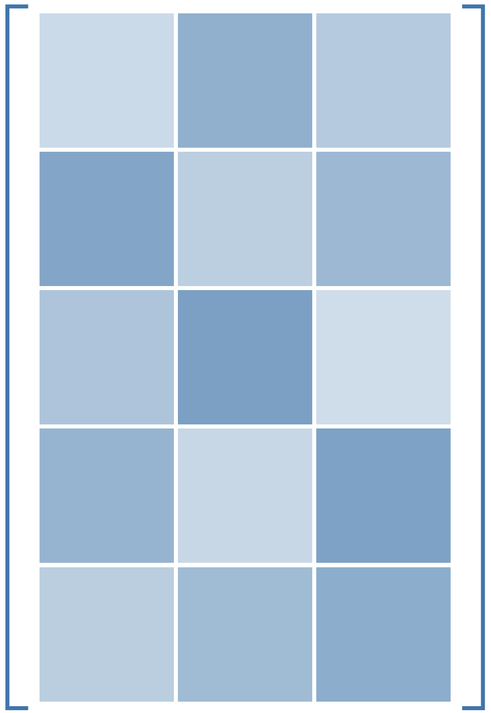
Historical Data H

Probabilistic Factor Model
Historical model factors $u_i \in \mathbb{R}^k$
Question factors $v_j \in \mathbb{R}^k$

$$P\left(H_{ij} = 1 \mid \{u_{i'}\}_{i'=1}^{N_{old}}, \{v_{j'}\}_{j'=1}^{N_q}\right) \approx \sigma(u_i^\top v_j)$$

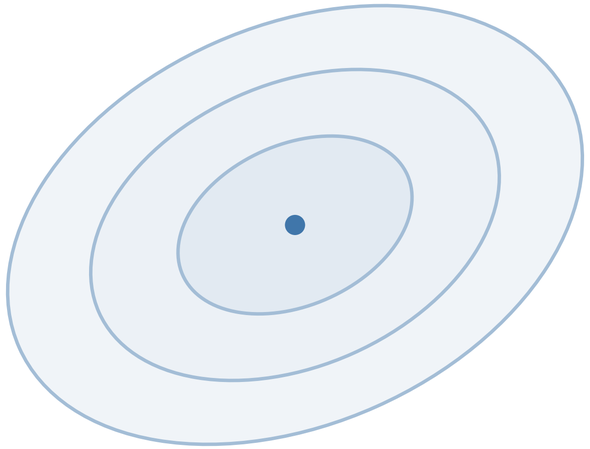
- Performance driven by small number of shared capabilities (e.g., coding, reasoning, and domain knowledge)
- Factor model only for efficiency gains. Coverage holds regardless of fidelity

1. Learn structure from history: initializing a new model factor



Historical Model Factors U

Empirical Mean
& Covariance

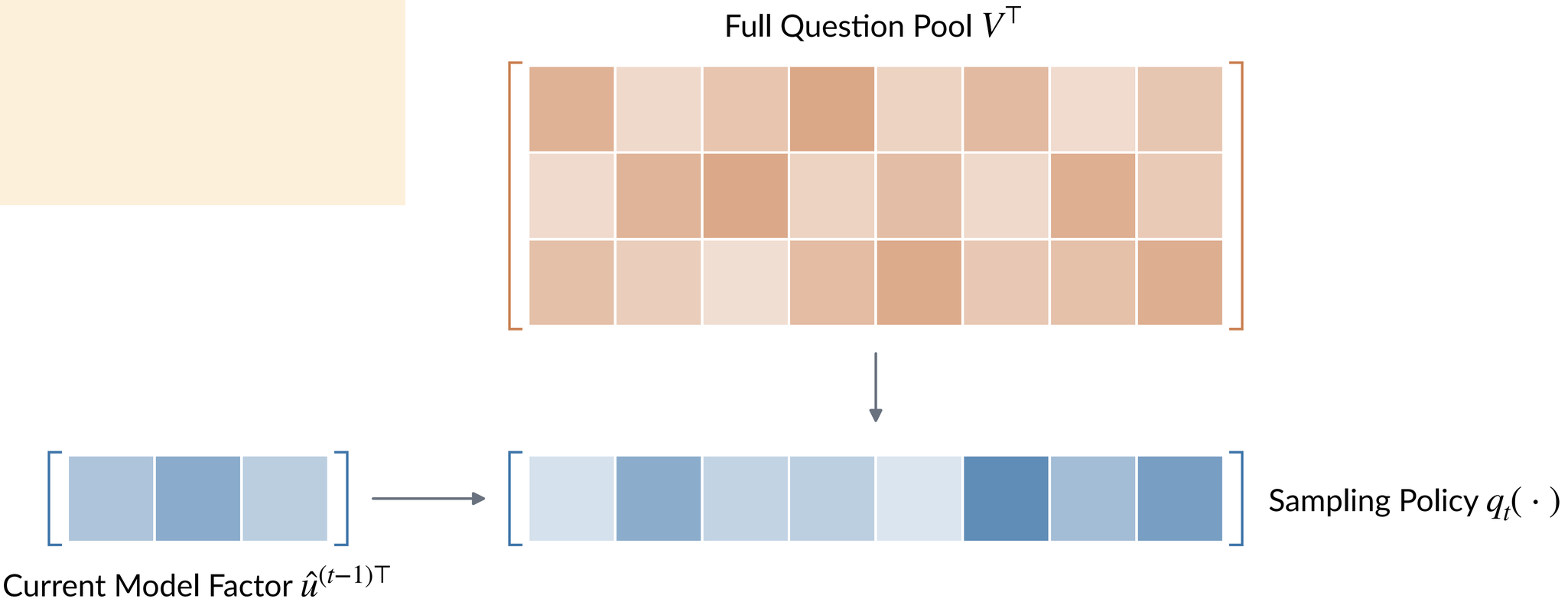


New Model Factor $u \sim \mathcal{N}\left(\hat{u}^{(0)}, \hat{\Sigma}^{(0)}\right)$

2. Adaptive querying: proactive selection + model updates.

Action Log:

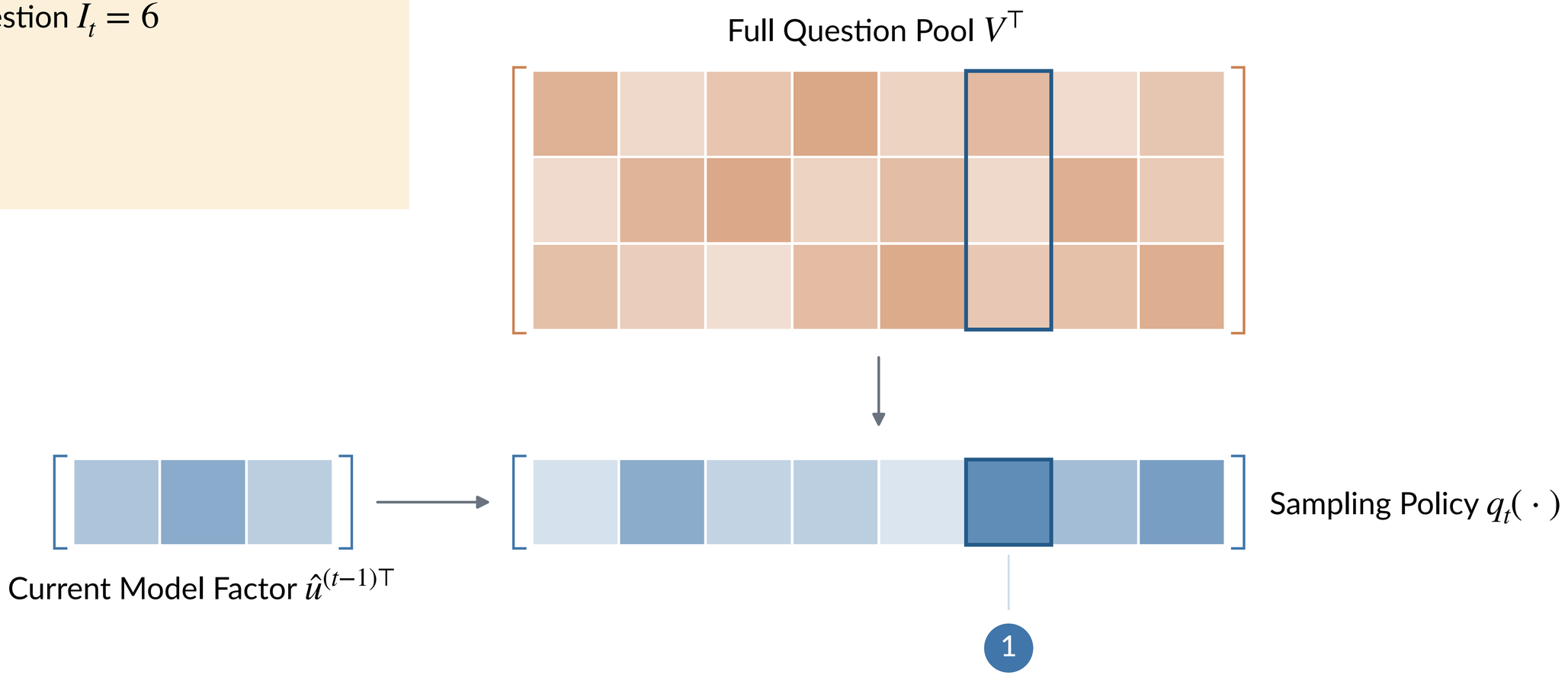
- Computed sampling policy $q_t(\cdot)$



2. Adaptive querying: proactive selection + model updates.

Action Log:

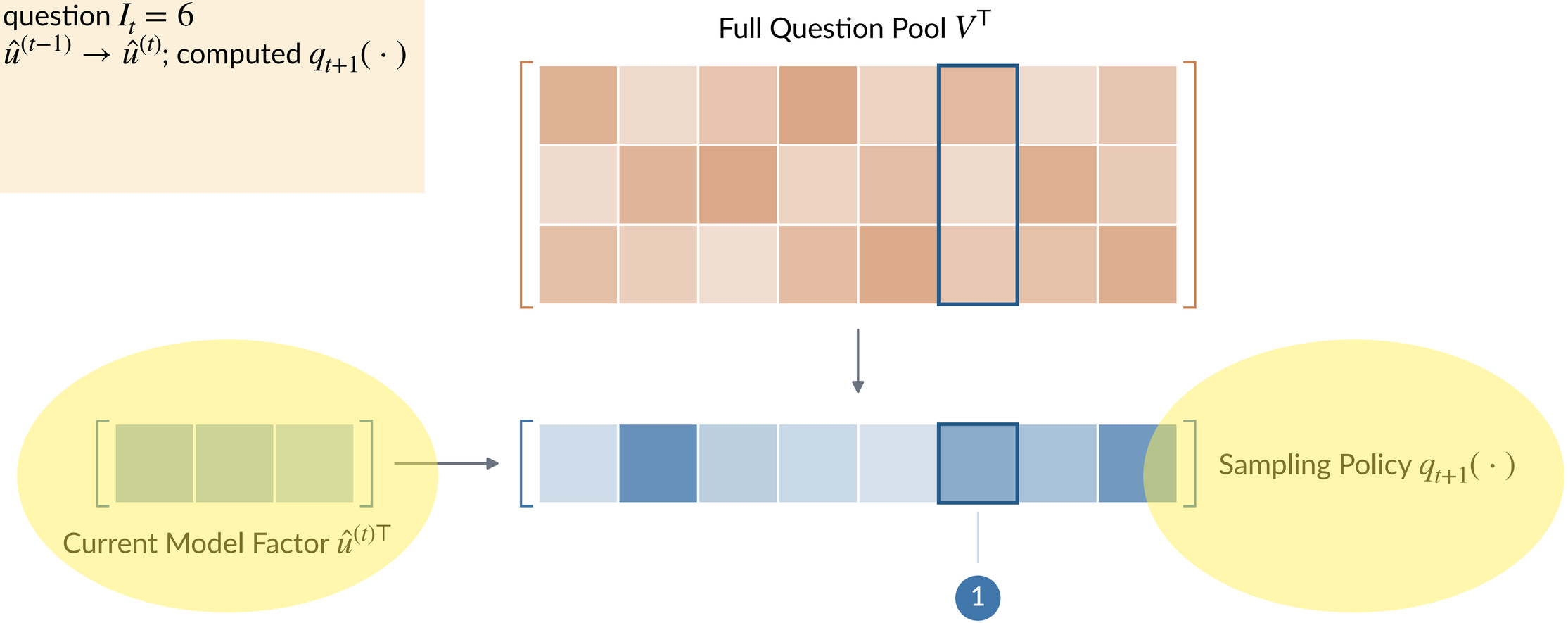
- Computed sampling policy $q_t(\cdot)$
- Selected question $I_t = 6$



2. Adaptive querying: proactive selection + model updates.

Action Log:

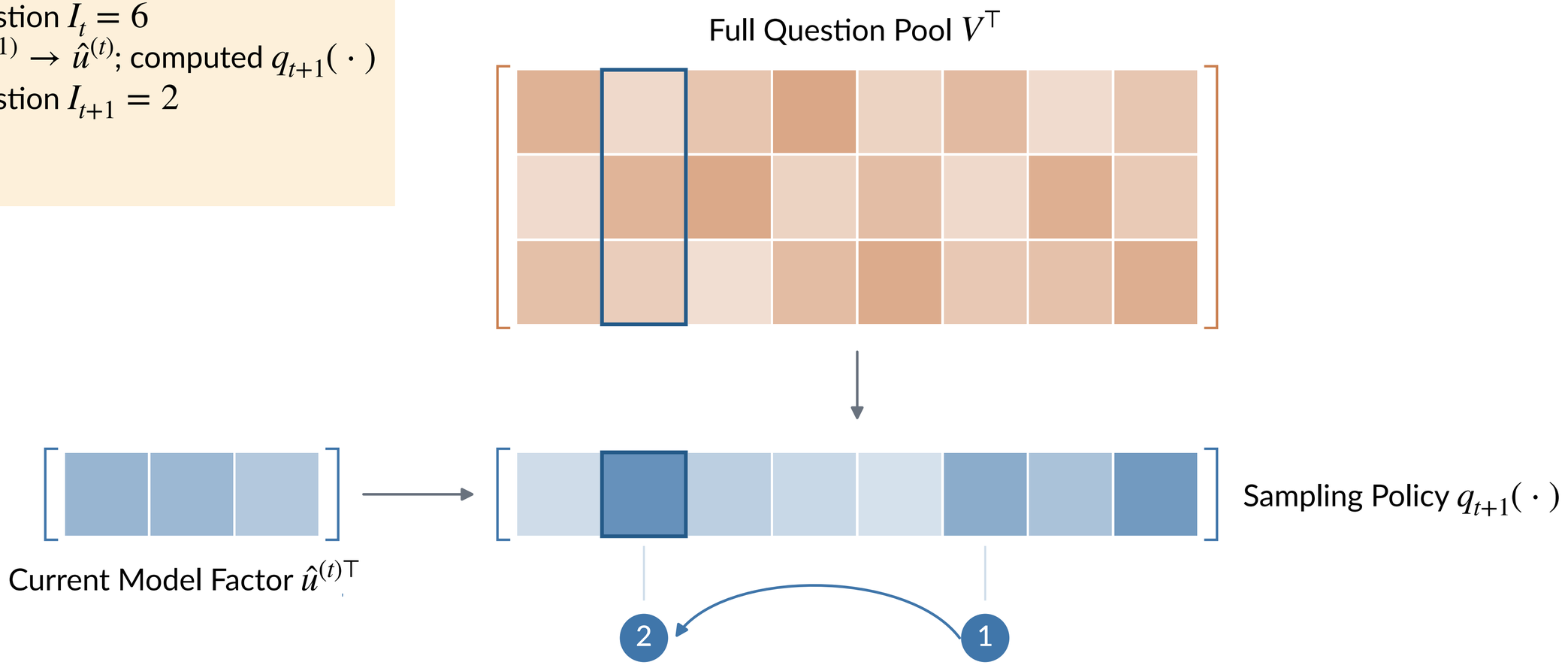
- Computed sampling policy $q_t(\cdot)$
- Selected question $I_t = 6$
- Updated $\hat{u}^{(t-1)} \rightarrow \hat{u}^{(t)}$; computed $q_{t+1}(\cdot)$



2. Adaptive querying: proactive selection + model updates.

Action Log:

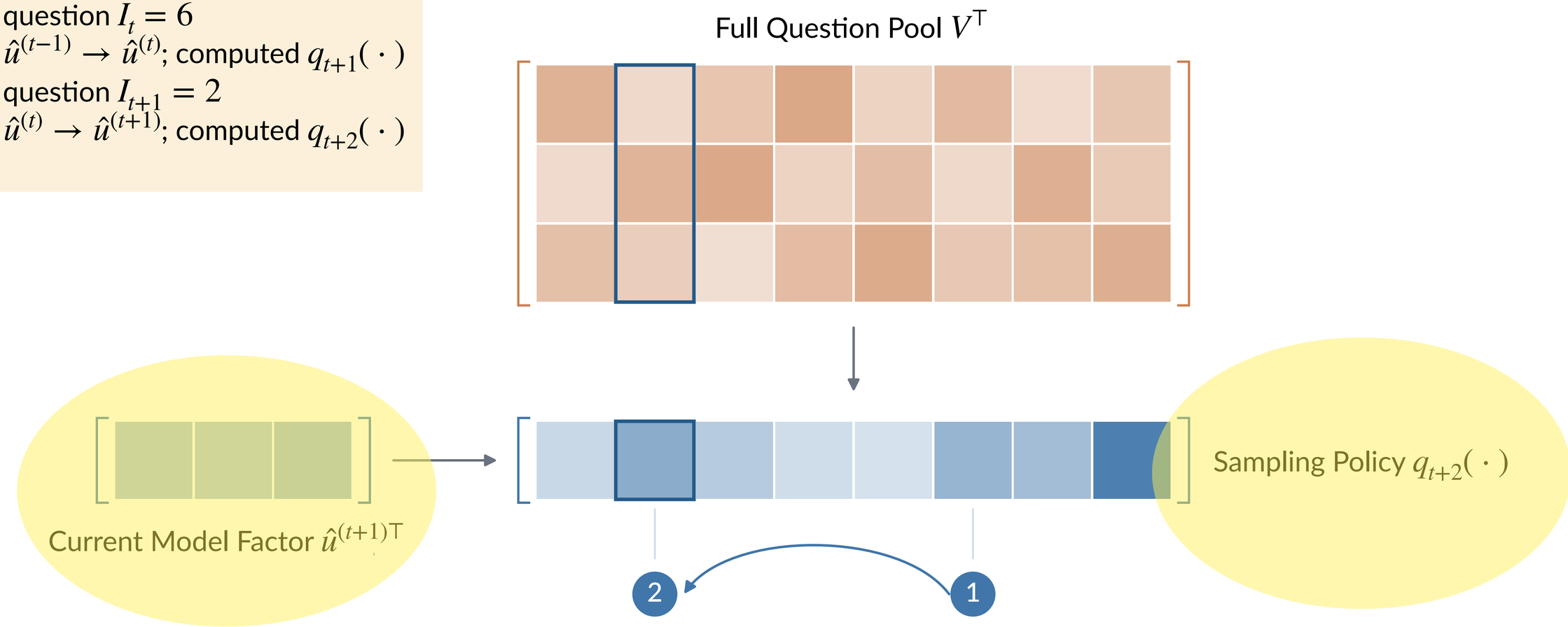
- Computed sampling policy $q_t(\cdot)$
- Selected question $I_t = 6$
- Updated $\hat{u}^{(t-1)} \rightarrow \hat{u}^{(t)}$; computed $q_{t+1}(\cdot)$
- Selected question $I_{t+1} = 2$



2. Adaptive querying: proactive selection + model updates.

Action Log:

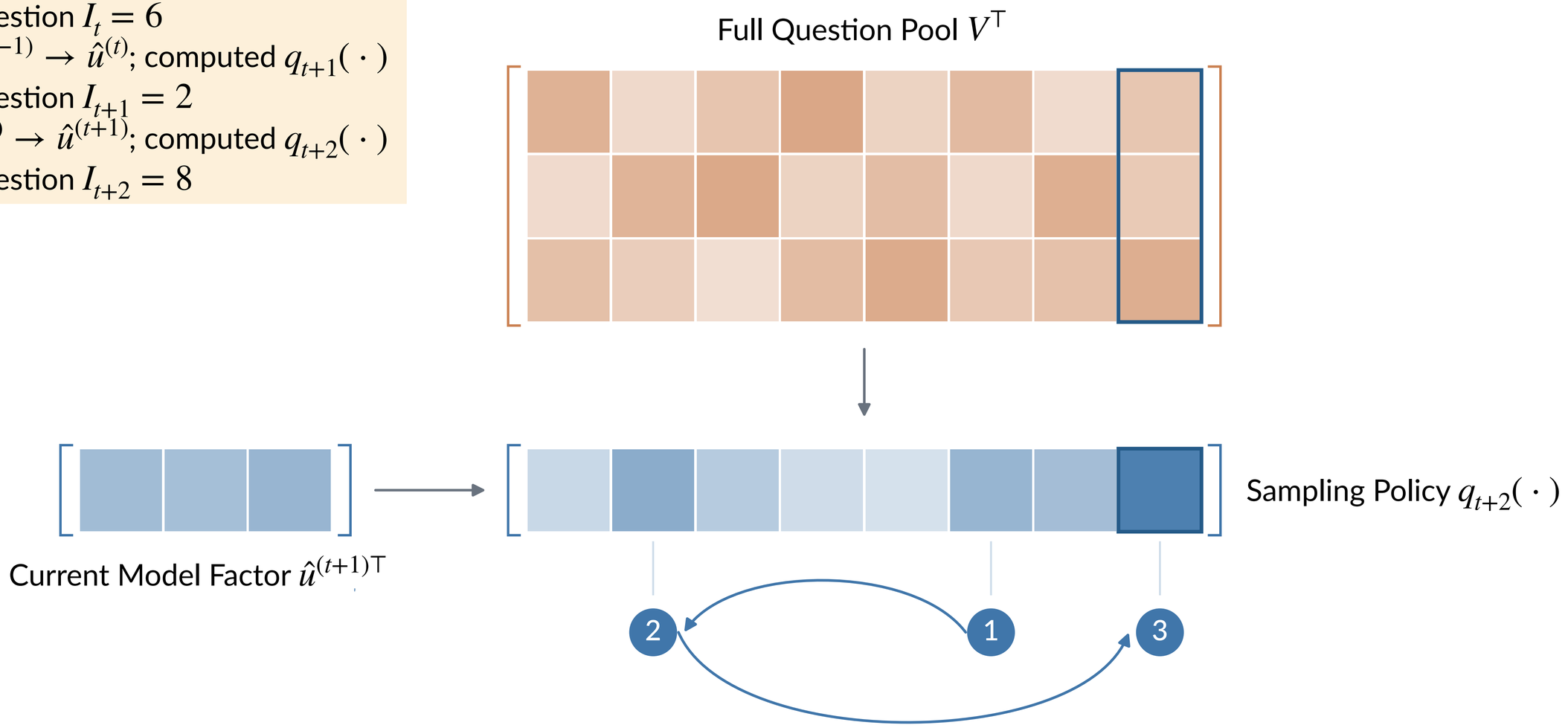
- Computed sampling policy $q_t(\cdot)$
- Selected question $I_t = 6$
- Updated $\hat{u}^{(t-1)} \to \hat{u}^{(t)}$; computed $q_{t+1}(\cdot)$
- Selected question $I_{t+1} = 2$
- Updated $\hat{u}^{(t)} \to \hat{u}^{(t+1)}$; computed $q_{t+2}(\cdot)$



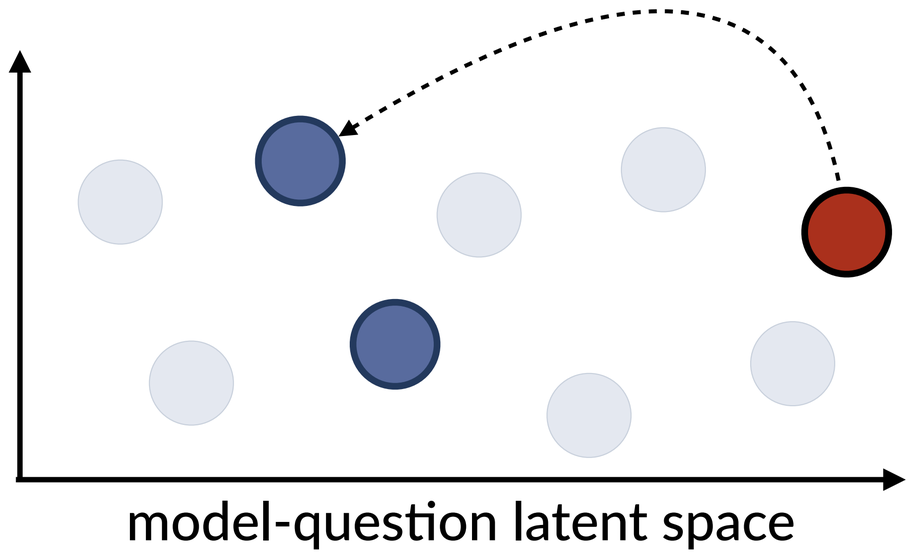
2. Adaptive querying: proactive selection + model updates.

Action Log:

- Computed sampling policy $q_t(\cdot)$
- Selected question $I_t = 6$
- Updated $\hat{u}^{(t-1)} \to \hat{u}^{(t)}$; computed $q_{t+1}(\cdot)$
- Selected question $I_{t+1} = 2$
- Updated $\hat{u}^{(t)} \to \hat{u}^{(t+1)}$; computed $q_{t+2}(\cdot)$
- Selected question $I_{t+2} = 8$



2. Adaptive querying: “explore, then exploit.”



Adaptive + Informative Sampling Policy:

Adaptive: pick next question based on all previous outcomes + most recent factor model predictions

Informative:

1. Bayesian Active Learning
(Exploration)

“My initialization for a new model’s factor is definitely not right. Which questions do I pick to quickly lock down this model in factor space?”

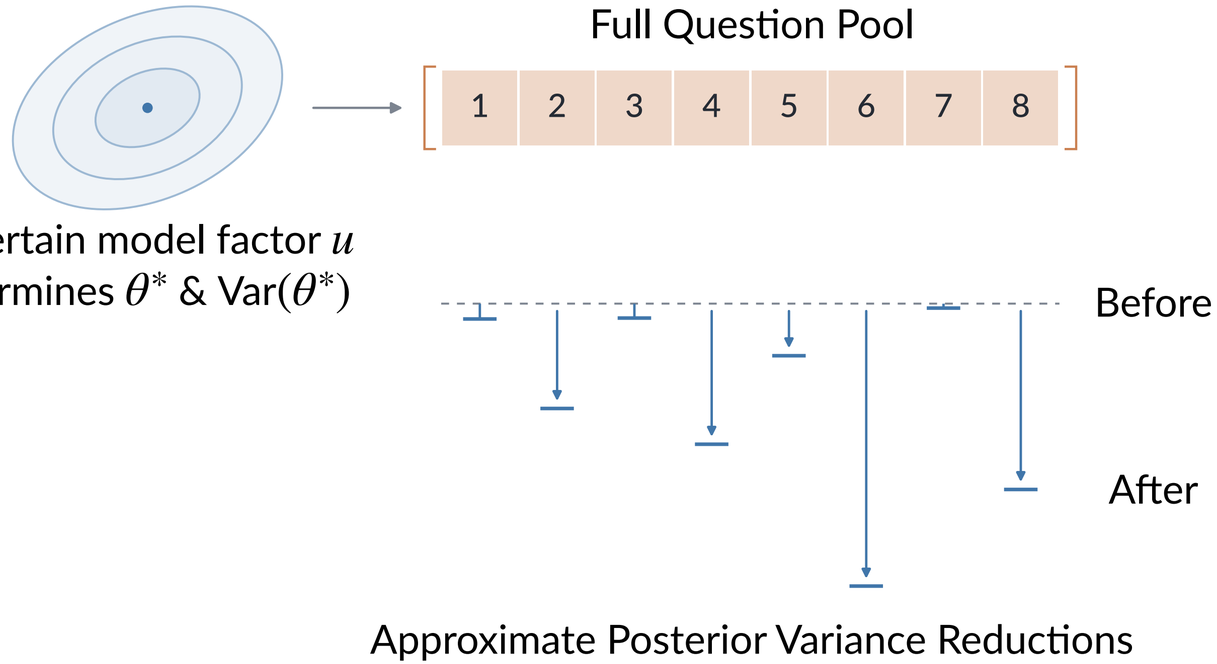
2. Neyman Allocation
(Exploitation)

“I’ve learned a good-enough new model factor now. Which benchmark question outcomes are implied to have the highest uncertainty under my factor model?”

2. Adaptive querying: exploration via Bayesian Active Learning

In Bayesian Factor Model World (Imaginary):

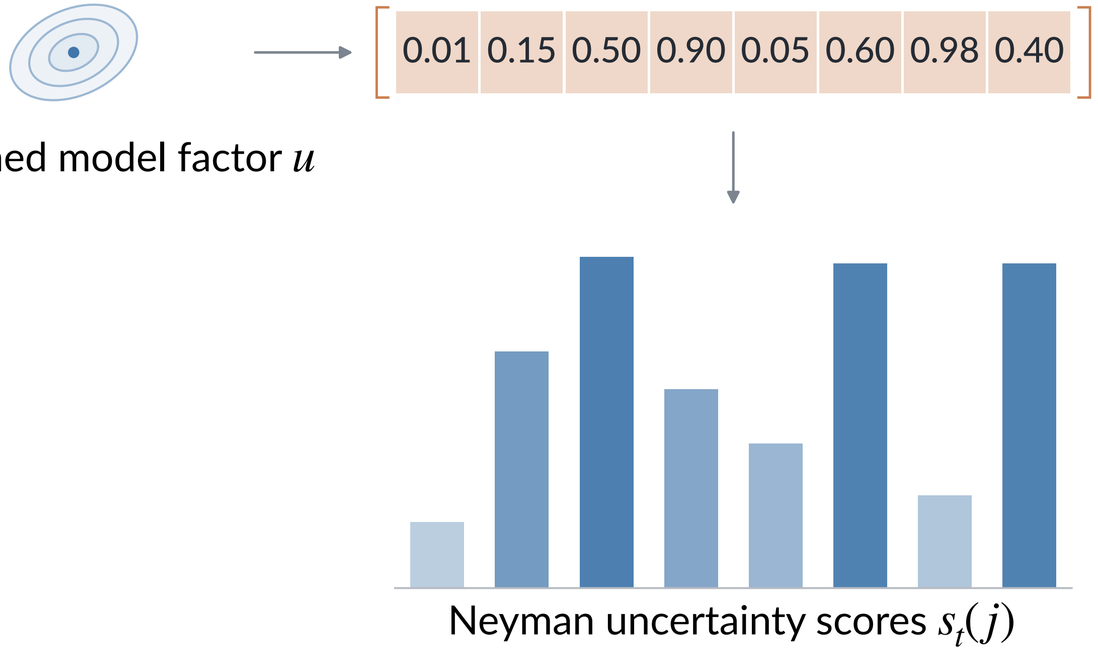
$$\theta^* := \frac{1}{N_q} \sum_{j=1}^{N_q} \sigma\left(u^\top v_j\right), \quad u \sim \mathcal{N}\left(\hat{u}^{(t-1)}, \hat{\Sigma}^{(t-1)}\right)$$



2. Adaptive querying: exploitation via Neyman allocation

Imagine our model factor u is "perfect." Which questions to sample?

$$\hat{p}_j^{(t-1)} := \sigma\left(\hat{u}^{(t-1)\top} v_j\right), \quad s_t(j) := \sqrt{\hat{p}_j^{(t-1)}\left(1 - \hat{p}_j^{(t-1)}\right)}$$



3. Reporting accuracy + CI: valid inference under adaptive querying

What we have:

- Adaptively queried n_b question outcomes
- Factor model predicting all questions' outcomes

Problem:

How do we combine everything together into one prediction with a statistically-valid CI?

Difficulty:

Adaptive data collection + ML predictions violate traditional statistical inference

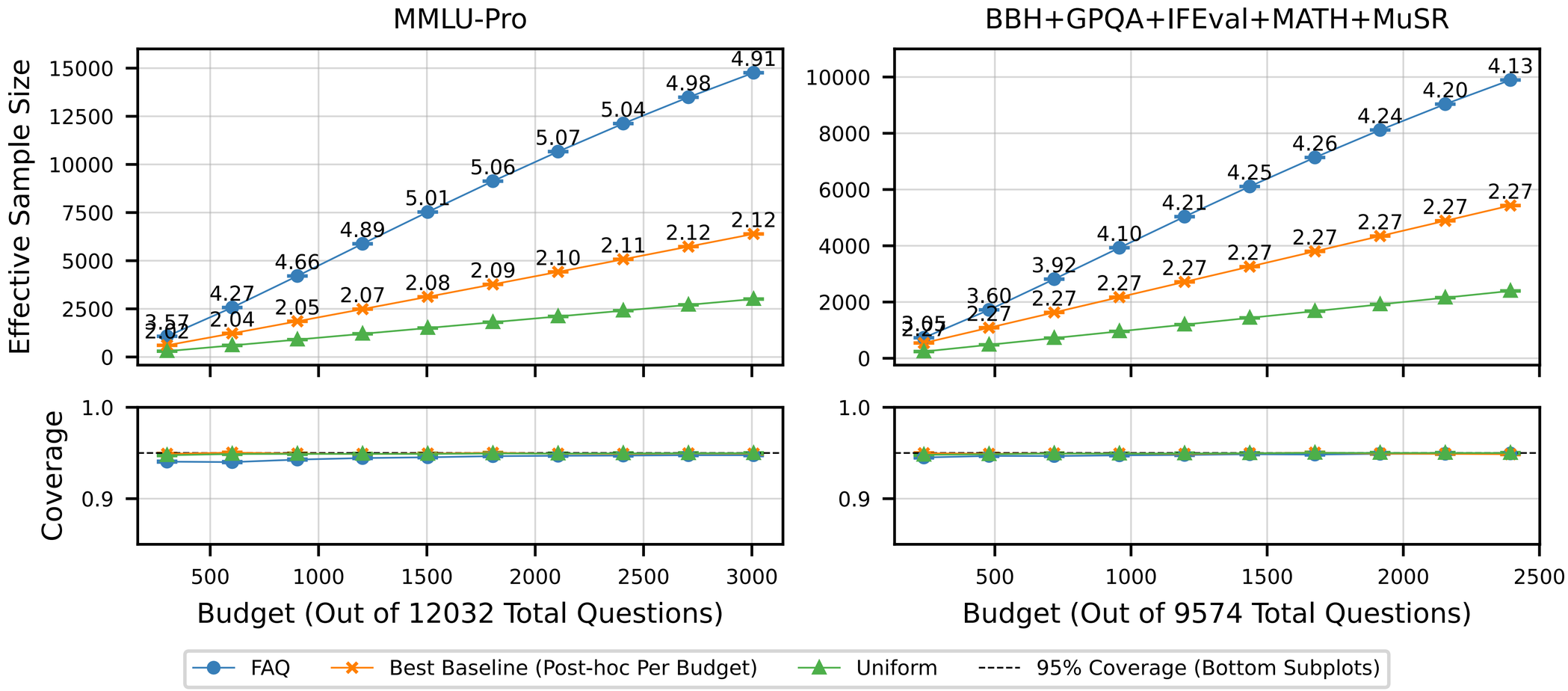
Solution: Pro-Active Inference (PAI)

Final PAI Accuracy Estimate:
(CI via Martingale CLT)

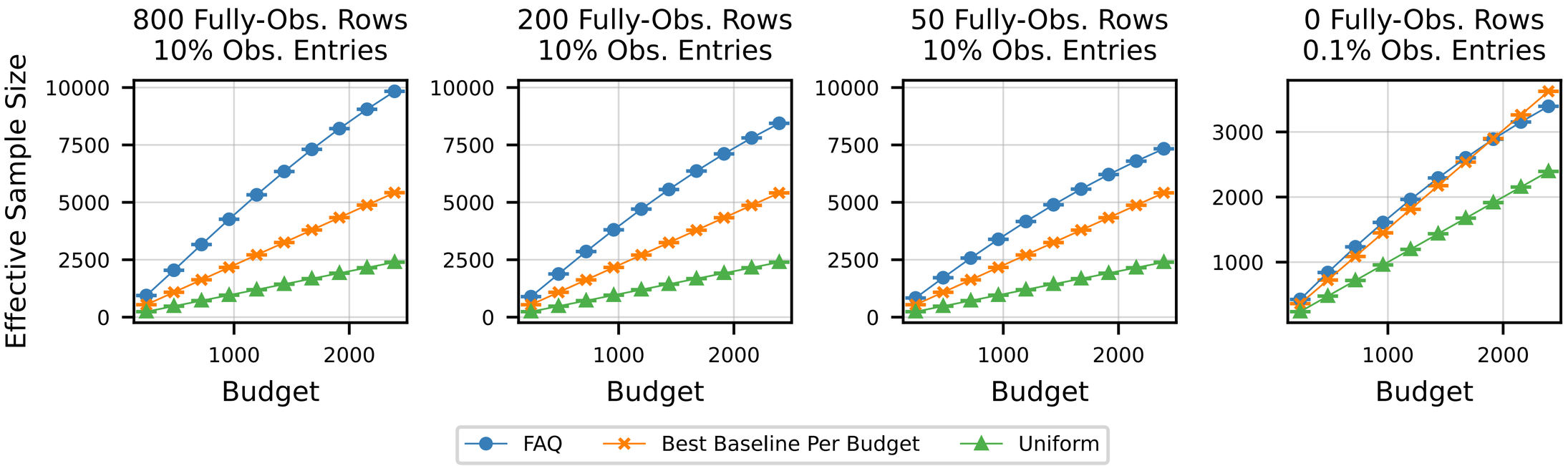
$$\hat{\theta}_{n_b} := \frac{1}{n_b}\sum_{t=1}^{n_b}\phi_t, \quad \phi_t := \underbrace{\frac{1}{N_q}\sum_{j=1}^{N_q}\hat{p}_j^{(t-1)}}_{\text{Factor Model Imputed Predictions}} + \underbrace{\frac{1}{N_q}\frac{z_{I_t}-\hat{p}_{I_t}^{(t-1)}}{q_t(I_t)}}_{\text{Debiasing Correction}}$$

*__Recall:__ correctness $z_j \in \{0,1\}$ on question j.

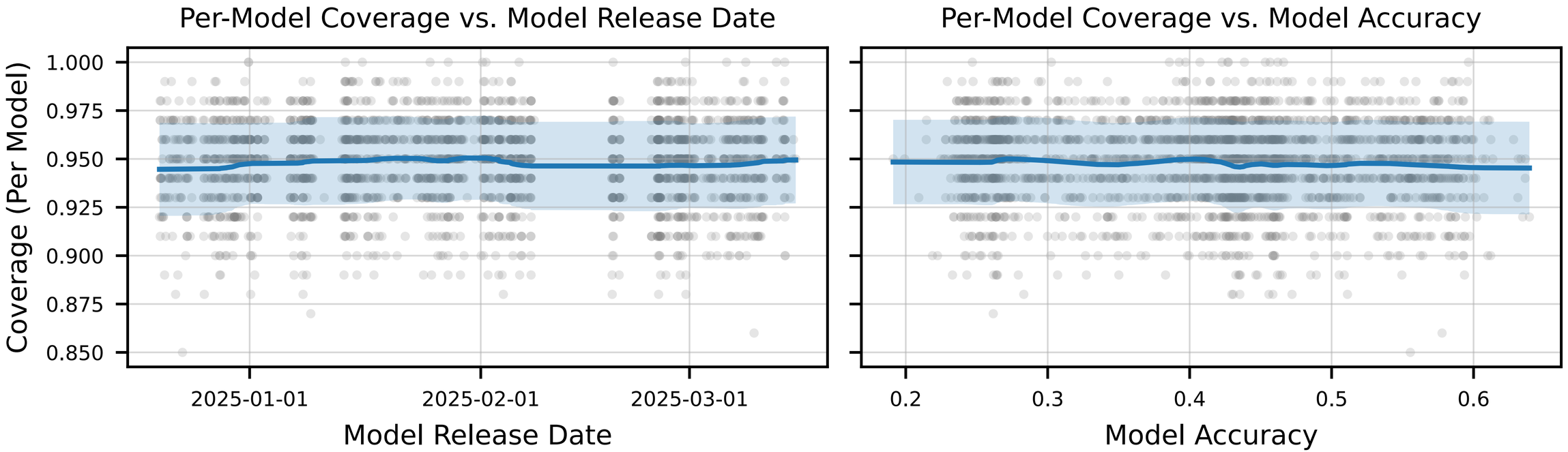
FAQ can achieve up to 5x higher effective sample size than uniform sampling



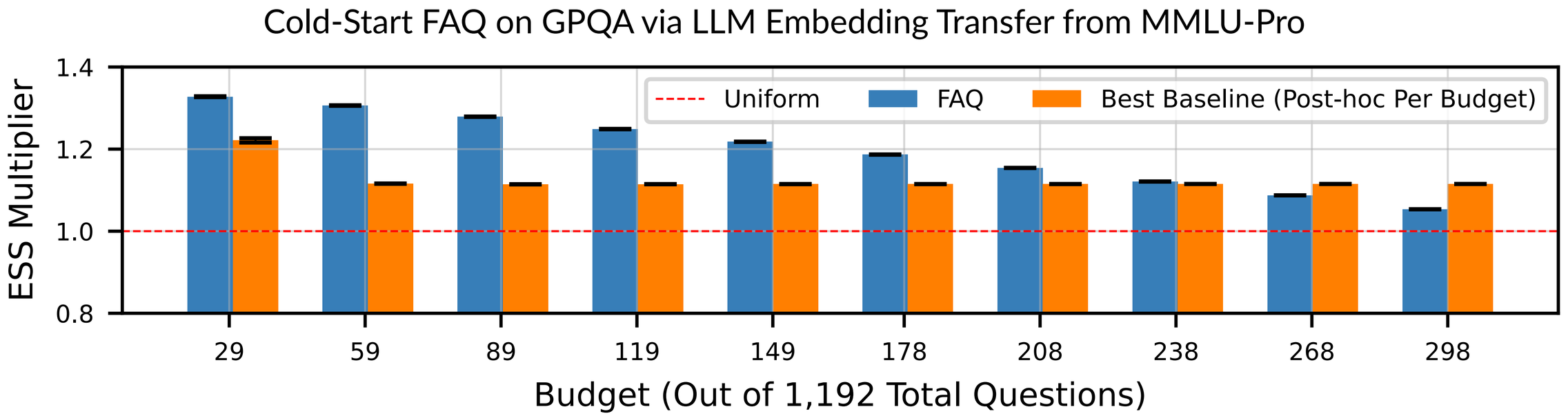
FAQ retains performance edge even under severe historical missingness



OK, but how does coverage actually look in the real world?



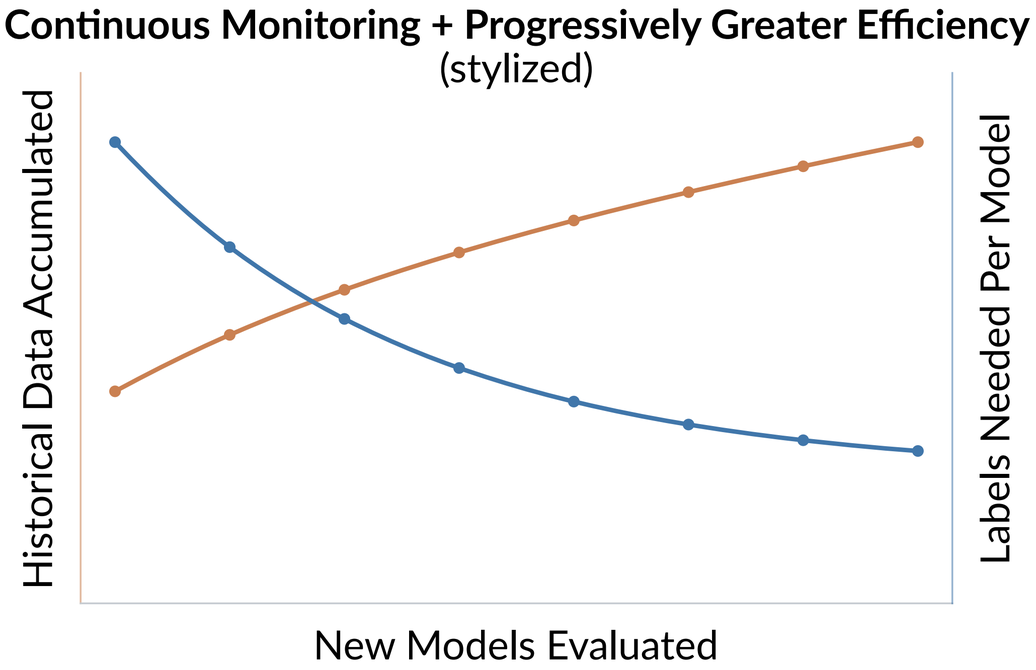
And what if I have no historical data for a benchmark?



*All post-hoc selected baselines were given 0.1% observed historical data from GPQA!

Summary + what's next?

- **Factorized Active Querying (FAQ):** query-efficient LLM evaluation with statistical guarantees
  - **Probabilistic Factor Model:** learning latent structure from partially-observed history
  - **Hybrid Sampling Policy:** Bayesian active learning ("explore") + Neyman allocation ("exploit")
  - **Pro-Active Inference:** valid statistical inference under nontraditional adaptive querying + ML updates
- **Continuous monitoring:** each evaluation leverages accumulated historical data while contributing fresh outcomes, increasing cost-efficiency over time
- **Fundamentally, a proof of concept:** much fancier surrogates + sampling policies are possible!
- **Additional evaluation paradigms:** non-binary feedback, intermediate reasoning traces, etc.



Thank you so much for your time!
I’m happy to take any questions.

Factorized Active Querying (FAQ)
[NeurIPS 2026]

Pro-Active Inference (PAI)
[arXiv]